AuroraStay

Policy RAG POC

Oct 11

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Executive Summary

- A reproducible local RAG baseline is built from four policy guides. It retrieves cited passages and runs all 20 supplied cases. Offline proxy scores are diagnostic only; hosted judging and inference remain unverified.

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Business Problem and Solution

- Staff reconcile policy details across four guides; 40% of 1,200 weekly queries are complex.
- Local retrieval POC returns grounded, page-cited evidence for staff review.

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Policy Data

- Four supplied guides cover booking, room/property facts, refunds/charges, and conduct/accessibility. Extraction produced 19 pages. Key risk: property, rate plan, timing, and exception details must not be generalized.

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Text Representation

- Embedding: corpus-fitted normalized TF-IDF vectors (local, deterministic; no pretrained max length). Similarity: cosine via normalized vectors. This is a lightweight POC baseline, not the domain-specific embedding model contemplated in the starter.

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Chunk Design

- Recursive word-window chunking: 190 words per chunk, 35-word overlap; each chunk retains source PDF and page. Smaller, policy-section-aware chunks are a recommended next retrieval improvement.

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Policy Chunk Example

- 42 chunks from 19 pages. Example: “ASH-CHI Loop lists a $50 per night pet fee, capped at $150 per stay.” Citation metadata is retained as source filename and page.

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FAISS Vector Index

- FAISS IndexFlatIP stores 42 normalized vectors. The persisted index and JSON metadata allow restart without re-reading PDFs. Vector dimension: 1,008 corpus terms.

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Retrieval Example

- Query: “What is the pet fee at ASH-CHI Loop?” Retrieval returns policy passages with source/page and similarity. Staff should verify the live property profile before quoting a fee.

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Answer Generation

- Configured hosted option: OpenAI-compatible gpt-4o-mini, temperature 0.0. No model key was available, so tested answers used an extractive local fallback; hosted generation is not verified.

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Grounding Rules

- Prompt policy: answer only from retrieved passages; preserve property/rate-plan qualifications; cite filename and page; do not invent fees or promises; state uncertainty and verify with the property.

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RAG Demonstration

- The local RAG route accepts a question, retrieves up to 3 passages, and returns a policy sentence plus citation. Fallback mode is explicitly labelled offline_extract.

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Evaluation Set

- Evaluation.csv contains 10 questions, expected answers, and reference context. Topics include check-in, cancellation, pet fees, smoking, service animals, refunds, and accessibility.

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Judge Model

- Rubric plan: DeepEval GEval with a separate gpt-4o judge. It was not run because an approved judge-model secret is not configured. No judge score is claimed.

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Evaluation Measures

- Faithfulness: claims supported by retrieval. Contextual recall: evidence completeness. Contextual precision: evidence relevance. Answer relevancy: response addresses question. Current reported values are lexical proxies, not GEval.

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Evaluation Results

- 10/10 evaluation questions returned an answer. Offline proxies: support 1.000; context recall 0.625; context precision 0.123; answer overlap 0.387. These are not semantic judge scores.

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Held-Out Test Set

- test.xlsx was converted to test.csv and includes 10 held-out cases across booking channel, member offers, occupancy, accessible rooms, parking, breakfast, rate changes, checkout, pets, and card holds.

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Test Results

- 10/10 test questions returned an answer. Offline proxies: support 1.000; context recall 0.733; context precision 0.122; answer overlap 0.465. Low precision/relevancy show retrieval and response quality need tuning.

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Persisted Backend Index

- Backend persists policy.faiss and store.json under backend/index. The API loads this index at startup and returns the cited retrieval evidence.

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Flask API

- Flask routes: GET /health; POST /v1/relevant_chunks; POST /v1/answer; POST /v1/answer_with_relevant_chunks. Invalid or empty queries return HTTP 400.

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Backend Dependencies

- Backend: Flask, Gunicorn, FAISS CPU, NumPy, PyMuPDF, Requests. Frontend: Streamlit and Requests. Dockerfiles and Compose are included with the supporting source.

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Backend Container

- Backend Dockerfile serves port 7860 with Gunicorn. The vector index is copied into the image. Docker build/runtime has not been executed in this hosted project.

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Streamlit Interface

- Streamlit supports a staff question, top-k and response mode, answer display, and expandable source passages. It warns staff not to enter card or sensitive medical details.

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Frontend Container

- Frontend Dockerfile serves Streamlit on port 8501; Compose connects it to the Flask service. This is a local deployment scaffold, not a Codespaces deployment record.

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GitHub Publication

- Not published: the GitHub integration card was dismissed. There is no repository URL or verified push. Backend/frontend files are available in the supporting project folder.

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Hosted Inference

- No public Codespaces URL or live inference screenshot exists. Do not submit a fabricated endpoint. This 6-point deployment-inference criterion remains unverified.

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Retrieved Evidence API

- Local API: POST /v1/relevant_chunks with {query, k}. Response includes retrieved content, source, page, and similarity. This validates local retrieval only, not forwarded online inference.

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Answer API

- Local API: POST /v1/answer_with_relevant_chunks. Response includes answer, mode, cited source chunks, and similarity. No hosted call was made.

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APPENDIX

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Completion and Submission Checklist

- Submit the presentation PDF only for the low-code path. Keep the notebook as working evidence. Before claiming full rubric completion: run DeepEval, publish files to GitHub, deploy both containers in Codespaces, capture live inference, and replace proxy results with judge scores.

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Thank You

